(a) Methodological Ground Simulation & Chamber Parity

Chamber 1: Geomagnetic Control (GMF)

Magnetic Regime: 45–50 μT Ambient Terrestrial Field

Experimental Purpose: Terrestrial environmental baseline

Chamber 2: Near-Null Field (NNMF)

Magnetic Regime: <10 nT (Shielded / Actively Cancelled)

Experimental Purpose: Lunar surface magnetic simulation

Chamber 3: Matched Bifilar Sham Control

Magnetic Regime: Opposing windings; net B = 0

Experimental Purpose: Controls for ohmic heat, vibration & EMI

(b) In Situ Causal Reversal Protocol (Proof of Concept)

Three-Step Causal Proof Inside Magnetic Shielding

Step 1: Hypomagnetic Exposure

<10 nT inside μ-metal shield

Elicits emergence lag, chlorosis & ROS leakage

Step 2: Vector Re-application

45 μT active vector restored in situ

Re-applied via internal triaxial Helmholtz coils

Step 3: Biological Phenotype Rescue

Phenotype restored to GMF baseline

Definitively proves direct magnetic causality

Definitively rules out microclimatic, ethylene, or enclosure artifacts

(c) Flight Payload Trade Space: Science Gaps vs. Engineering Realities

- Active triaxial Helmholtz cancellation / bias coil hardware
- Heavy copper coil mass, high continuous power draw & Joule heat
- Severe spacecraft electromagnetic interference (EMI) certification
- Complex in-flight closed-loop vector control systems
- Best reserved for terrestrial laboratory analogues & ground testing

Exceeds Typical Flight Payload Resource Budgets (Ground-Only):

- Passive 3-axis magnetic sensing (miniature fluxgate or AMR)
- Canopy-level environmental vector logging (Bx, By, Bz, |B|)
- Minimal resource budget: < 1 W electrical power, grams mass
- Continuous logging captures spacecraft DC avionics & stray fields
- Enables post-flight correlative science & phenotype attribution

Realistic Flight Payload Accommodation (Recommended):

(d) Post-Flight Phenotypic Disambiguation Framework

Differentiating Environmental Stressors in Lunar Agricultural Payloads

Near-Null Magnetic Field (<10 nT)

- Emergence lag (Δt50 = +12–36 h); capacity drop 10–30%
- Repressed chlorophyll (HEMA1, POR) & anthocyanins (CHS)
- Mitochondrial ROS leak (AOX1D, CAT3); intact nuclear DNA
- Diagnostic: Rescued by 45 μT vector; insensitive to 1g spin

Lunar Partial Gravity (0.166 g)

- Normal germination capacity; minimal emergence lag
- Directional skewing, random root waving, cell wall remodeling
- Normal plastid greening; apoplastic ROS (RBOHD/F)
- Diagnostic: Rescued by 1g centrifuge; insensitive to B-field

Deep-Space Ionizing Radiation (GCR)

- Emergence failure only at lethal doses; apical arrest
- Severe DNA double-strand breaks, γ-H2AX foci & ATM/ATR
- Widespread systemic oxidative damage & lipid peroxidation
- Diagnostic: Mitigated by shielding; correlated with dosimeter